Magnetism in Rocks – Teacher Copy

Authors: Kira Lee, University of Tasmania

In this class, students quantify the magnetic field strength of different rocks/ minerals using the phyphox magnetometer on a smart device, and then explore crustal magnetic anomalies using the Geoscience Australia portal.

Earth & space science curriculum link:

Compare the observable properties of soils, rocks and minerals and investigate why they are important Earth resources (AC9S3U02).

Use equipment to observe, measure and record data with reasonable precision, using digital tools as appropriate (AC9S5I03 AC9S6I03).

Pre-requisite knowledge:

- Students should have a basic understanding of the three rock types (sedimentary, igneous and metamorphic) and know that a rock's physical properties depend on its composition/ mineralogy.
- Students should have basic familiarity with the phyphox app (see the document *Downloading and using the phyphox magnetometer*).

Aim:

- Calculate the magnetic field strength of different rocks/ minerals using a smart device magnetometer.
- Understand that the magnetic properties of a rock depends on the quantity of magnetic minerals it contains.
- Understand how remnant magnetism in crustal rocks contributes to Earth's total magnetic field, and how geomagnetic surveying is used to identify magnetic anomalies.

Materials:

- Smart device (phone/ tablet) with the phyphox app downloaded & access to the internet
- Steel nail
- Around 3 rock/ mineral samples no larger than hand-sized (ideally a sample of magnetite, an igneous rock and a sedimentary rock) (See **Figure A**)

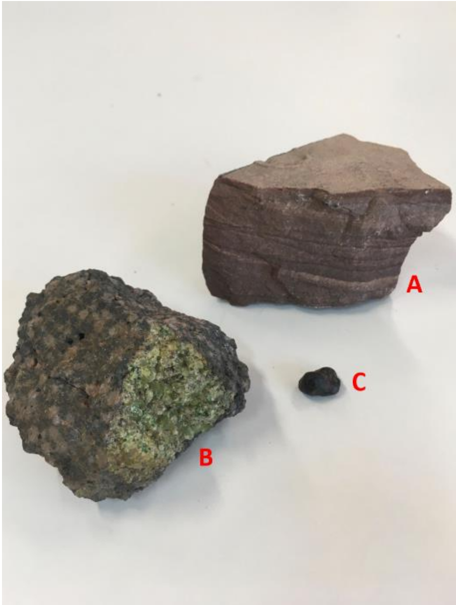

Figure A: Example rock/ mineral samples.

A: Sedimentary rock (sandstone); B: Igneous rock (peridotite); C: Magnetite

Safety considerations:

- Handle smart devices with care to avoid damage. Do not press down hard onto device screens with the steel nails!
- Remind students to handle sharp objects (nails) carefully to avoid injury to other students/ staff.

Activity 1: Predicting the magnetic properties of your samples

Using your existing understanding of magnetism and rocks, order your rock/ mineral samples from most to least magnetic.

Students should consider the mineral composition of their samples and rank them from most to least magnetic based on the amount of magnetic minerals (magnetite) they contain.

In the example of magnetite, peridotite and sandstone samples, the correct ranking would be:

(Most → least magnetic)

Magnetite → Peridotite → Sandstone

1. Which sample do you think is the <u>most</u> magnetic? Why?

Students should predict that the magnetite (or their most magnetite-rich sample) is the most magnetic because the more magnetite a sample contains, the stronger its magnetic field.

2. Which sample do you think is the <u>least</u> magnetic? Why?

Students should predict that their sedimentary rock sample is the least magnetic because it contains no/ only very small amounts of magnetic minerals.

Activity 2: Measuring the magnetic field strength of your samples using the phyphox app

We will now quantify the magnetic field strength of your samples using a magnetometer app on your smart device.

Note: You will need to have the phyphox app downloaded on your device for this activity. See the document *'Downloading and using the phyphox magnetometer'* for guidance.

<u>Step 1: Locating the Magnetic Sensor in your Smart Device</u>

Demonstration video: Magnetism in Rocks Step 1: Locating the Magnetic Sensor in your Smart Device

Before we begin collecting geomagnetic data, it is important to know where the magnetometer sensor in your smart device is located.

You have been given a steel nail.

1. Place your device on a flat surface. Do not move it during the experiment.
2. Begin recording data with the phyphox magnetometer in **Graph** mode. Allow the readings to stabilise for 5-10 seconds.
3. Slowly move the nail across your device screen from bottom to top (be careful not to scratch your screen!).

4. The readings on all 3 graphs will spike when the nail is close to the sensor (see **Figure 1**).

In newer iPhone devices (iPhone X and onwards), this should be the top left corner.

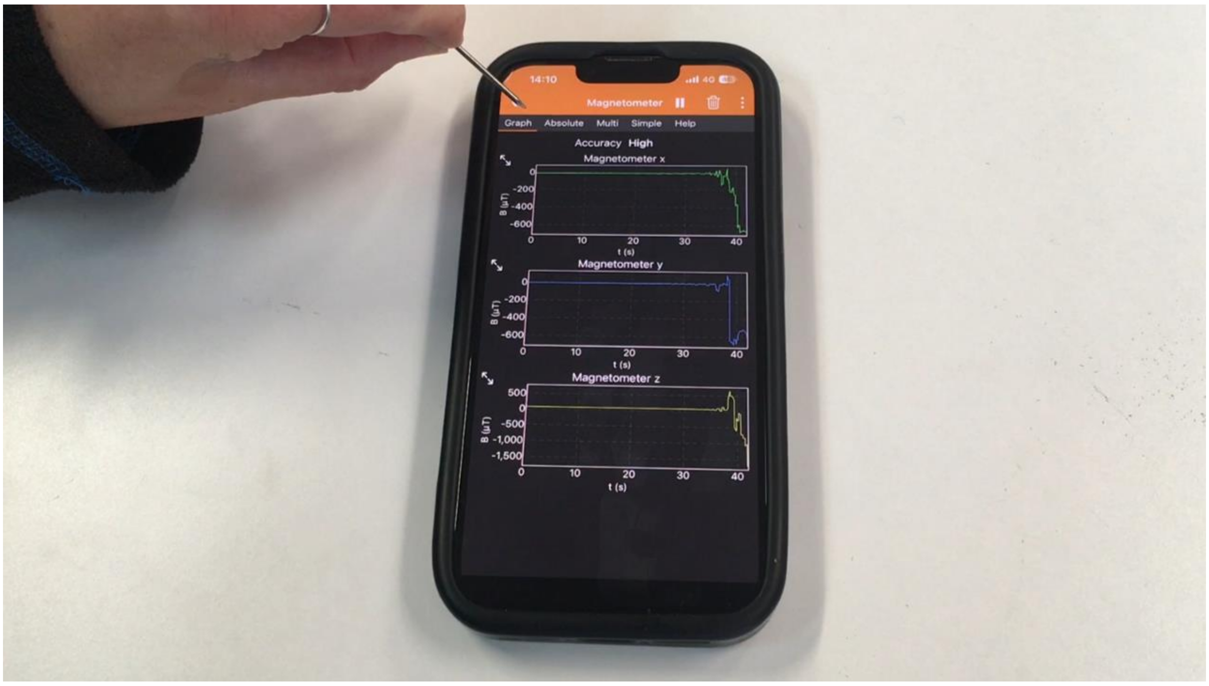

Figure 1: phyphox magnetometer readings spike when the nail is close to the phone's magnetometer sensor (top left corner in this iPhone 13).

Step 2: Measuring the Magnetic Field Strength of Rocks using Phyphox

Demonstration video: [Magnetism in Rocks Step 2: Measuring the Magnetic Field Strength of Rocks using Phyphox]

1. Place the device on a table away from metal/ electronic objects. DO NOT MOVE the phone from this position during the experiment.
2. Begin recording using the **Simple** magnetometer mode in phyphox. Allow the readings to stabilise for 5-10 seconds.
3. Then record the **Absolute** magnetic value – this is **Earth's ambient magnetic field (A).**

Note: the absolute value is the total magnetic field and is calculated from the x, y and z components using the Pythagorean theorem $B_{Absolute} = \sqrt{B_x^2 + B_y^2 + B_z^2}$

4. Leave your phone in its current position. Now place your rock/ mineral sample close to/ over the magnetic sensor in your device.

5. Allow the new readings to stabilise for 10 seconds then record the new Absolute magnetic value – this is the **total magnetic field (T).**

Note: the readings may change rapidly if your sample has a strong magnetic field. Hold the sample in one position over the magnetometer sensor and record the average value as best you can.

6. Now we will isolate the **magnetic field of your sample (S)** by subtracting the value of Earth's ambient magnetic field (A) from the total magnetic field (T). An example is provided below in **Table 1**.

Table 1: Example magnetic measurements for a rock sample.

Rock/Sample name	Ambient magnetic field (A) /μT	Total magnetic field (T) /μT	Sample magnetic field (S) (= T – A) /μT
Example	64	72	72 – 64 = 8

If the value of S is greater than ±0.5 μT, phyphox has detected that your sample has a magnetic field.

Repeat this process for each of your samples.

Recalibrate the device's magnetometer before measuring each sample to reset the readings. To do this, move it in a sweeping figure-8 motion as shown in the demonstration video.

The samples that are the most magnetic will have a higher total field strength.

Rank your samples again from most to least magnetic based on the total field strengths you calculated. Was your prediction in **Activity 1** correct?

Remind students to:

- Not move their device whilst they are measuring a sample as this will change the readings.
- Use the phyphox magnetometer in **Simple** mode – this displays the magnetometer values numerically on the screen instead of as graphs.
- Recalibrate their device between measuring samples (see demonstration video) to reset their magnetometer readings, especially after measuring magnetite/ strongly magnetic samples as these can saturate the sensor.
- Record new ambient field strength readings before measuring each sample.

The values displayed on the magnetometer change depending on the orientation of the sample and the distance between the sample and the device's sensor. They can change rapidly if the sample is strongly magnetic.

The absolute magnetic field strength values students record do not have to be a perfect average – the relative differences in readings between samples is the most important concept.

The sign (+ or -) of the change in the magnetic field depends on the positioning of the sample over the magnetometer sensor (which has + and - axes) and the direction of magnetism within the sample (which can be + or -). Students shouldn't worry about this, the magnitude of the change in the magnetic field is more important than the sign.

Expected results for the total magnetic field strength of a magnetite sample, an igneous sample and a sedimentary sample (sandstone) are shown in the **Table A**.

Values are given as a broad range to reflect how the μT value of a rock isn't a fixed quantity, and depends on sample geometry, orientation, distance to the sensor.

Table A: Expected student results.

Sample	Expected Sample Magnetic Field (μT)
Magnetite	100+
Igneous Rock	1 – 10
Sedimentary Rock	0 – 1

From their calculations, students should verify that the more magnetite-rich samples have the greatest magnetic field strength. They should rank them again from most to least magnetic and see if their prediction in **Activity 1** was correct.

Activity 3: An introduction to magnetic anomalies and geomagnetic surveying

Navigate to the Geoscience Australia Portal: https://portal.ga.gov.au/

What is Geoscience Australia?

Geoscience Australia (GA) is a government agency that conducts geoscientific research. They advise the government on everything related to geoscience, and store geological and geographical data collected by the commonwealth (Wikipedia, 2026).

The portal opens a map of Australia.

Select the 'Layers' option from the toolbox and in the search bar type 'Total Magnetic Intensity'.

Select **'Magnetic Intensity Image'** from the list and click 'Add to Map'.

Your map should look like the following image (**Figure 2**):

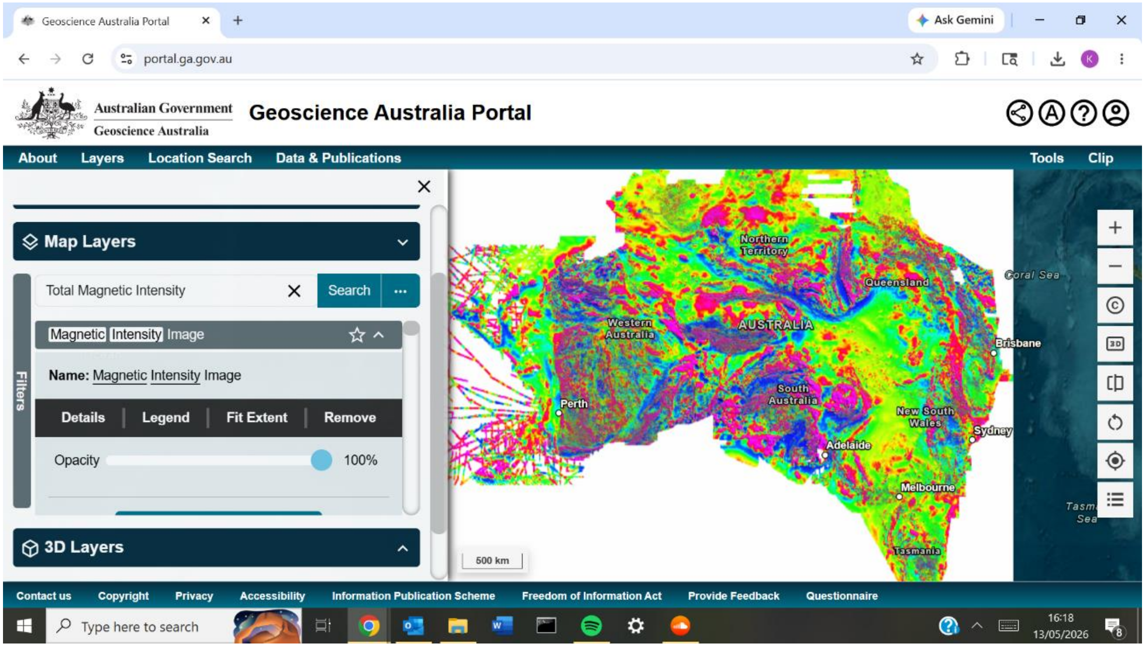

Figure 2: Total Magnetic Intensity (TMI) image of Australia (Minty and Poudjom Djomani, 2019). Viewed via Geoscience Australia Portal (https://portal.ga.gov.au/).

Information about this dataset can be found here.

The map shows the magnetic field strength of crustal rocks. This was measured by magnetometers during geomagnetic surveying. The scale is in nanotesla (nT) (1 nT = 0.001 μT).

Magnetic anomalies are variations in the geomagnetic field due to the differing magnetic properties of rocks in the lithosphere (Luyendyk, 2025).

- Strong magnetic anomalies are coloured bright red/ pink or a deep blue. These are areas of strongly magnetic rocks.

- Weak magnetic anomalies are coloured green or yellow. These are areas of weak or non-magnetic rocks.

Explore the map using the zoom and pan controls.

- Zoom using the + or - buttons on the right of the screen, the scroll wheel on a mouse, or 'pinch to zoom' if your device is touchscreen.
- Pan by clicking and dragging the screen or swiping across the screen on a touchscreen device.

Based on your study of the map and your experimentation with rock samples in **Activity 2**, answer the following questions:

1. What might strong magnetic anomalies indicate about the geology of an area?

The area contains igneous rocks (like basalt, peridotite or dolerite). These are rich in iron-rich minerals like magnetite so are strongly magnetic.

2. What might weak magnetic anomalies indicate about the geology of an area?

The area contains sedimentary rocks (like sandstone) that do not contain magnetic minerals so aren't magnetic/ are only weakly magnetic.

HINT: Navigate to Hobart and zoom in until you can see the bright red/ pink rounded shapes to the west of the city – this is the Wellington Range. The Wellington Range is primarily composed of **dolerite** – an igneous rock rich in the magnetic mineral magnetite. These dolerites are strongly magnetic, so appear as a strong magnetic anomaly on the map.

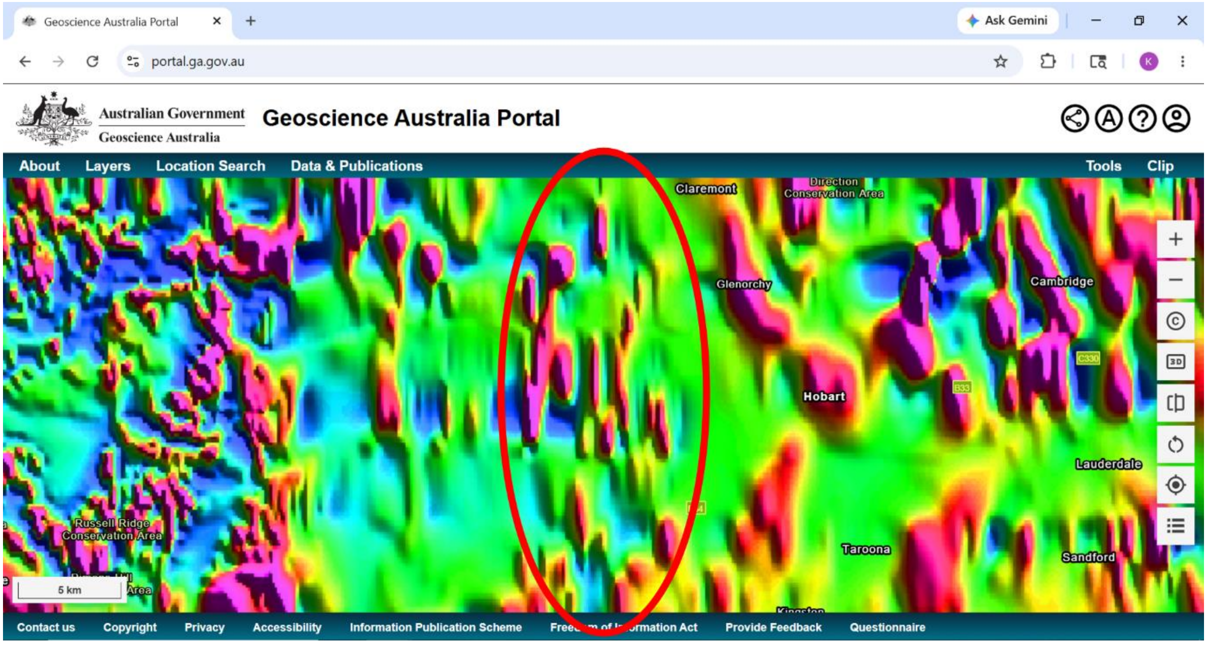

Figure B: Total Magnetic Intensity (TMI) image of Australia (Minty and Poudjom Djomani, 2019). Viewed via Geoscience Australia Portal (https://portal.ga.gov.au/).

The Wellington Range is circled in red.

Students should attempt to identify the Wellington Range on the map. This is circled in red on **Figure B.** The bright red/ pink parallel rounded shapes are likely the dolerite peaks.

By identifying the Wellington Range, seeing that it is coloured red/ pink, and being told that the rocks are magnetite-rich dolerite, students should make the connection that strong magnetic anomalies = strongly magnetic igneous rocks. This should help them answer the questions in Activity 3.

This activity was inspired by:

NASA (2025). *Exploratory Activities in Magnetism with Smart Devices*. Available at: https://science.nasa.gov/learn/heat/resource/exploratory-experiments-in-magnetism-with-smart-devices/

NASA (2025). *Introductory Experiments in Magnetism with Smart Devices*. Available at: https://science.nasa.gov/learn/heat/resource/introductory-experiments-in-magnetism-with-smart-devices/

This activity uses the phyphox mobile app: phyphox (Physical Phone Experiments) app, 2nd Institute of Physics, RWTH Aachen University.

References:

Luyendyk, B. P. (2025). Oceanic crust. *Encyclopedia Britannica*. Available at: https://www.britannica.com/science/oceanic-crust

Minty, B.R.S. and Poudjom Djomani, Y. (2019). Total Magnetic Intensity (TMI) grid of Australia 2019: seventh edition, 40 m cell size. Geoscience Australia, Canberra. Available at: https://doi.org/10.26186/5dd5fb557fffb (Accessed: 14 May 2026). Viewed via Geoscience Australia Portal.

Wikipedia (2026). *Geoscience Australia, Wikipedia*. Available at: https://en.wikipedia.org/wiki/Geoscience_Australia (Accessed: 15 June 2026).

This resource was developed with assistance from ChatGPT (OpenAI, 2026), used to generate and refine instructional ideas.